

2026 iPSC from JAX record

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2026年3月17日 星期二

The order details are below.

 **JAX_Order_5843769_PO.pdf**

Thaw the cells in 1*6-well for each with 2mL Stemflex+ROCKI

Table1						
	A	B	C	D	E	F
1		short name for the cell line	Details	sample ID on tube	Passage #	ID when I thaw cells
2	control	iPSC85	JIPSC1000, JIPSC1000_KOLF2.1J_human iPSC	2114953285	P3	3
3	TDP43 Q331K	iPSC60	JIPSC1066, JIPSC1066_TARDBP_Q331K_SNV/WT_human iPSC	3525175760	P6	2
4	TDP43 M337V	iPSC52	JIPSC1108, JIPSC1108_TARDBP_M337V_SNV/WT_human iPSC	3525176352	P6	1

2026年3月18日 星期三

Cells attached. 85 died a lot, but still have enough cells there.

Change media to the one without ROCKI

2026年3月20日 星期五

Change media

2026年3月21日 星期六

Split cells. each line (one well of 6-well plate) into 3*10-cm with ReleSR

Since I broke iPSC85 into too small aggregates with P1000, I added ROCKi into the plate.

2026年3月22日 星期日

Cells looked great.

Change media for iPSC85.

2026年3月23日 星期一

~40%

Change media for iPSC52 and iPSC60

2026年3月24日 星期二

~60%

Change media for iPSC85

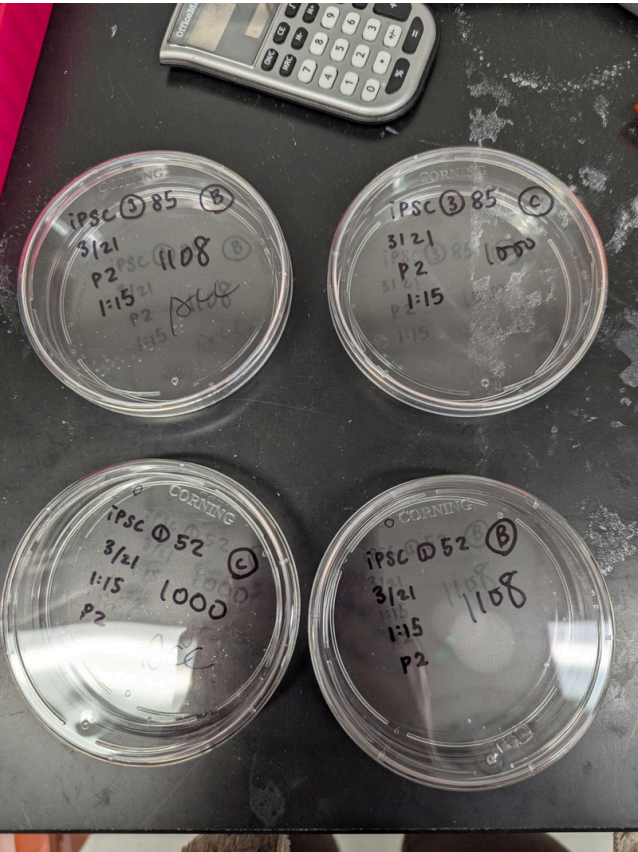
2026年3月25日 星期三

Freeze cells with ReLeSR*1 and accutase*2

20260325_freeze cells										
	A	B	C	D	E	F	G	H	I	J
1		Details	sample ID on tube	Passage #	"accutase 1M"	"accutase 1/5 10cm"		"ReLeSR 1/4 10cm"		
2	control	JIPSC1000, JIPSC1000_KOLF2.1J_human iPSC	2114953285	P5	cYST79-84	cYST0074-0078	<- this is the label on the tube, but i might swap them with 1108	cYST_0070-0073		
3	TDP43 Q331K	JIPSC1066, JIPSC1066_TARDBP_Q331K_SNV/WT_human iPSC	3525175760	P8		cYST0107-0116		cYST0101-0104		
4	TDP43 M337V	JIPSC1108, JIPSC1108_TARDBP_M337V_SNV/WT_human iPSC	3525176352	P8	cYST85-88, 95-100	cYST0090-0094	<- this is the label on the tube, but i might swap them with 1000	cYST0089, 0117-0119		

- I did it in three batches separately
- (labeled with "accutase 1M")** iPSC1000 (iPSC85) and iPSC1108 (iPSC52) with accutase -> freeze 1M/vial (6 vials for iPSC1000 and 10 vials for iPSC1108) (thin vials). For 6 vials for iPSC1000, I did not add ROCK inhibitor, but the rest of the vials I added 10uM ROCK inhibitor. The ROCK inhibitoer for freezing is not in the protocol, but I noticed it helps survival after thawing. (ref benchling note)
 - (labled with "accutase 1/5 10cm")** iPSC1066 (iPSC60)*2, iPSC1000*1 (I accidentally used iPSC52 instead of iPSC85, see picture below), iPSC1108*1 (I accidentally used iPSC85 instead of iPSC52, see picture below). (thin vials) --> just ordered the PCR and sequencing primer to validate the genotype. [JiPSC1108](#) & [JiPSC1066](#) (oYST77-82)
 - (labeled with "ReLeSR 1/4 10cm")** Use ReLeSR for all. Incubate for 30s-1min and aspirate ReLeSR. Incubate until 5min and flush every plate with 2mL Stemflex. Seed 100uL/one well of 6-well*2. Prepare 4.8mL + 1.2mL DMSO (20%DMSO). Add 2mL 20%DMSO of Stemflex into each line, pipette with 5mL pipet and distribute 1mL into each vial (used thick vials).

 PXL_20260326_001812367.jpg



2026年3月27日 星期五

Change a half of media

2026年3月28日 星期六

Split cells 1:6 (1 well to a whole plate of 6-well plate)

Table2		
	A	B
1		Passage #
2	iPSC1000	P3+4
3	iPSC1066	P6+4
4	iPSC1108	P6+4

2026年3月30日 星期一

~30% confluent. iPSC1066 and 1108 have a big cluster in the middle again...

2026年3月31日 星期二

50% confluent. Aside from the big cluster in the middle, there is still a lot of space around. --> decided to start differentiation next week.

2026年4月1日 星期三

Start differentiation (Cells are around 70-80% confluent)
Passage one well 1:3 and 1:6 to 2 wells of 6-well.

Table3			^
	A	B	
1		Passage #	
2	iPSC1000	P3+5	
3	iPSC1066	P6+5	
4	iPSC1108	P6+5	

2026年4月3日 星期五

Split the "1:3" cells (70% confluent) 1:10 into 4 wells of 6-well each.

Table4			^
	A	B	
1		Passage #	
2	iPSC1000	P3+6	
3	iPSC1066	P6+6	
4	iPSC1108	P6+6	

2026年4月4日 星期六

Change media ~20-30% confluent

2026年4月6日 星期一

~40-50% confluent. Some wells are ~30%
--> collect more confluent 3 wells and differentiate them in 25cm flask for suspension (Nunc™ Non-treated Flasks 169900)

2026年4月20日 星期一

Use all cells for differentiation
Extract gDNA for some cells from each line --> check PCR

2026年4月28日 星期二

gDNA PCR
oYST078 & oYST079: M337V
oYST081 & oYST082: Q331K
Sanger sequencing primer: oYST080 (M337V_1108) and oYST082 (Q331K_1066)

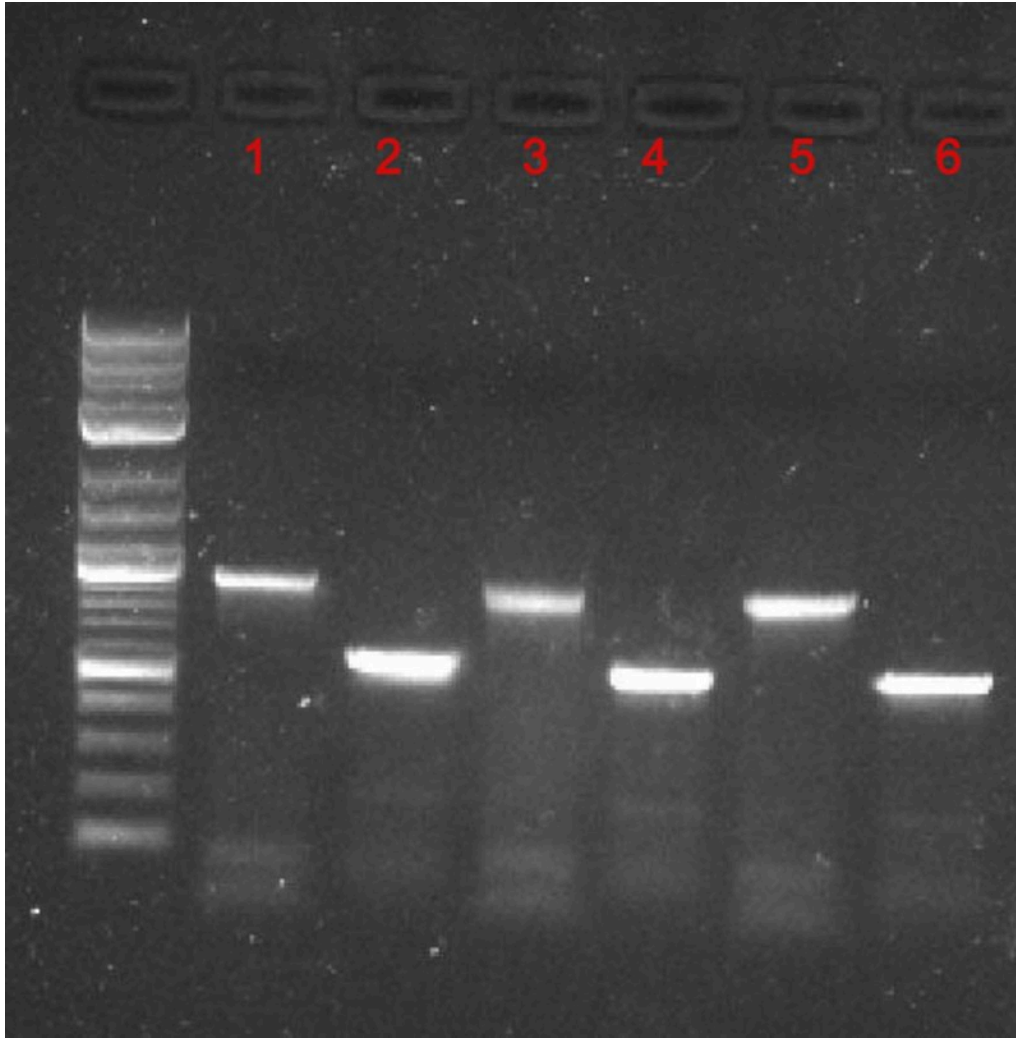
Table5						
	A	B	C	D	E	F
1	condition	cells	primer set	seq result		
2	1	1000	78+79	WT		
3	2	1000	81+82	WT	the seq shows a point mutation, but I think it's the error from the readout	
4	3	1066	78+79			
5	4	1066	81+82	Q331K	Bottom strand is T/G half and half (C->A)	
6	5	1108	78+79	M337V	A/G half and half	
7	6	1108	81+82			
8	7	H2O	78+79			
9	8	H2O	81+82			

PCR reagents1					
	A	B	C	D	E
1	sample #	4		plasmid	master mix
2	Template DNA 10 pg - 1 µg		0.3	2	
3	DreamTaq Green PCR Master Mix (2X)	25	10	10	45
4	Forward primer (100uM stock)	0.1-1.0 µM	0.2	0.2	0.9
5	Reverse primer (100uM stock)	0.1-1.0 µM	0.2	0.2	0.9
6	Water, nuclease-free	to 50 µL	9.3	7.6	41.85
7	total	50	20	20	

PCR for stock plasmid1				^
	A	B	C	
1	temp	time	cycle	
2	95	2min	1	
3	95	30s	35	
4	56	30s		
5	72	1min		
6	72	5min	1	

Gel check

767d8894-faf6-4bbe-b2e0-3e29b204275d.jpeg



2026年4月29日 星期三

PCR clean up and send it to sequencing

sample submission guide:

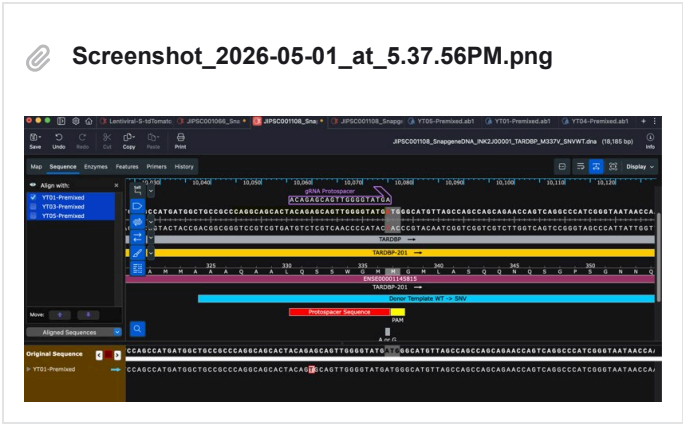
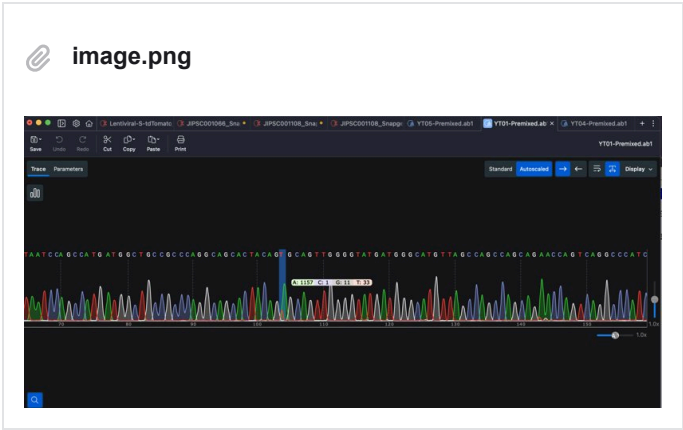
- 20ng DNA in 10uL
- 25pmol primer: 100uM*0.25uL
 - 1 uL of 100uM in 20uL -> add 5uL into each tube (3 tubes per primer: 80 for set 78+79, 82 for 81+82)

Sequencing results:

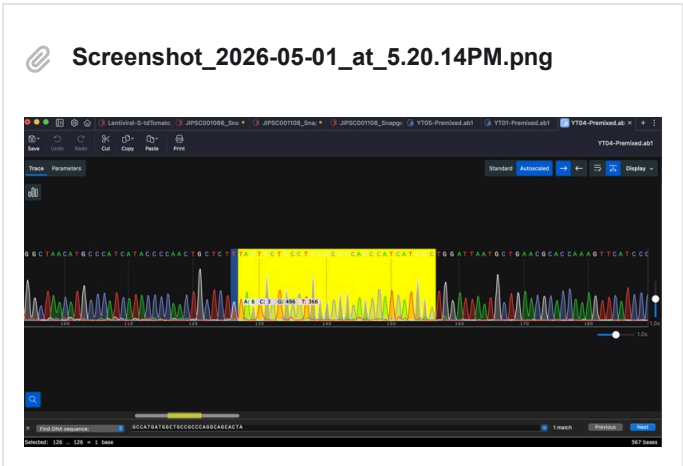
need to check .ab files since it's heterozygous

YT01

The T in summary data is actually A.



YT04 (Q331K, half T and half G)



YT05 (M337V, half A and half G)

